

**Design Thinking Project Report on
Crowd Sourced Civics Issue Reporting
and Resolution System**

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ABSTRACT

Rapid urbanization has led to a significant rise in civic problems such as potholes, garbage accumulation, water leakage, and damaged infrastructure. Traditional complaint management systems are often slow, inefficient, and lack transparency, resulting in delayed responses and reduced citizen satisfaction.

The Crowd-Sourced Civics Issue Reporting and Resolution System is a web- or mobile-based platform designed to empower citizens to actively participate in governance by reporting civic issues in real time. Users can submit complaints along with images, location details, and descriptions, ensuring accurate and actionable information for authorities.

The system leverages crowdsourcing techniques where multiple users can validate, comment on, or upvote reported issues, improving reliability and prioritization. Advanced technologies such as Artificial Intelligence (AI) and Machine Learning (ML) may be integrated to automatically classify issues, verify authenticity, and assign priority levels based on severity and urgency.

Furthermore, the platform provides real-time tracking, geolocation mapping, and analytics dashboards for authorities, enabling efficient decision-making and faster resolution of complaints. By enhancing transparency, reducing manual effort, and encouraging citizen engagement, the system bridges the gap between the public and government, contributing to smarter and more responsive urban governance.

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1. INTRODUCTION

In today's rapidly growing urban environments, cities face numerous civic challenges such as poor waste management, road damage, water leakage, streetlight failures, and sanitation issues. These problems directly affect the quality of life of citizens and require timely attention from municipal authorities. However, conventional complaint systems are often manual, time-consuming, and lack proper tracking mechanisms, leading to delays, inefficiency, and a lack of accountability.

With the widespread use of smartphones and internet connectivity, there is a growing opportunity to involve citizens directly in governance processes. A Crowd-Sourced Civics Issue Reporting and Resolution System leverages this potential by enabling individuals to report civic problems instantly through a digital platform. Citizens can upload images, provide descriptions, and share location details of issues, making the reporting process more accurate and transparent.

This system is based on the concept of crowdsourcing, where community participation plays a crucial role in identifying and validating problems. Multiple users can view, comment, and support reported issues, helping authorities prioritize them based on urgency and public concern. Such collaboration not only increases efficiency but also builds a sense of responsibility among citizens.

Additionally, the integration of modern technologies like geolocation services, cloud computing, and Artificial Intelligence (AI) enhances the system's capability to categorize issues, detect duplicates, and monitor resolution progress. Authorities can use dashboards and analytics to make informed decisions and allocate resources effectively.

Overall, this system acts as a bridge between citizens and government bodies, promoting transparency, accountability, and faster resolution of civic issues. It supports the vision of smart cities by encouraging active public participation and improving urban management through technology.

1.1 EXISTING SYSTEM

- The current system is mostly manual (phone calls, office visits, written complaints), making it time-consuming and inconvenient.
- It lacks transparency, as citizens cannot easily track the status of their complaints after submission.
- There is slow response and no proper prioritization of issues, leading to delays in resolving important civic problems.

1.2 PROPOSED SYSTEM

- A digital platform (web/mobile) that allows citizens to report civic issues instantly with images, location, and descriptions.
- Provides real-time tracking and transparent status updates, ensuring accountability and faster resolution by authorities.

2. SYSTEM ANALYSIS

2.1 SOFTWARE REQUIREMENTS

Operating System:

- Windows 10 or later
- macOS 10.15 or later
- Linux distributions with kernel version 4.15 or later
- iOS 13 or later
- Android 8.0 (Oreo) or later

Software:

- Updated web browser (for web-based interface)
- GPS and location services enabled
- Bluetooth enabled (for real-time tracking and indoor navigation)

Network:

- Internet connection for initial setup, map downloads, and updates
- Offline functionality supported through cached data.

2.2 HARDWARE REQUIREMENTS

Processor:

- Minimum: Dual-core CPU (2.0 GHz or faster)
- Recommended: Quad-core CPU (2.5 GHz or faster)

Memory (RAM):

- Minimum: 2 GB
- Recommended: 4 GB or more

Storage:

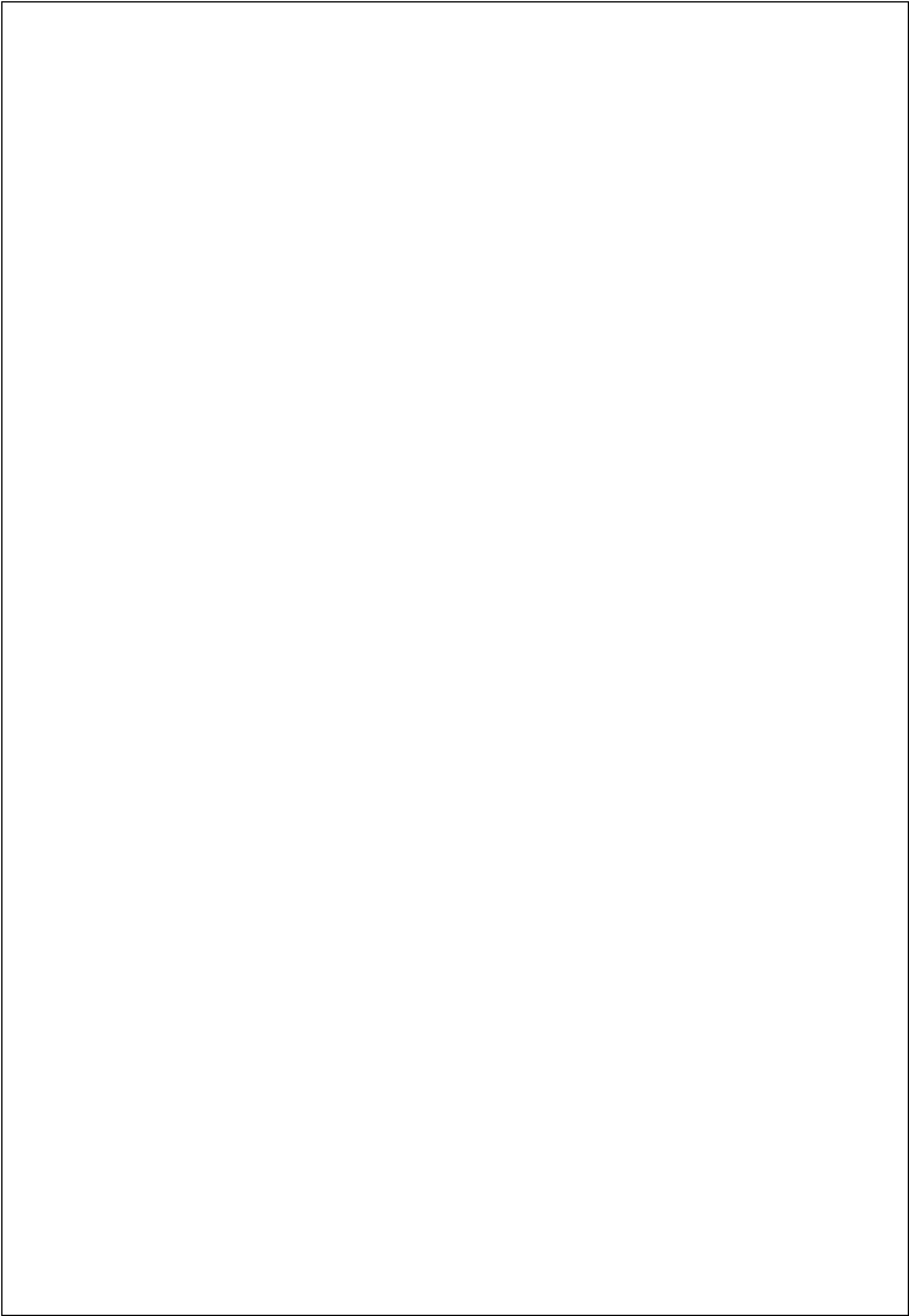
- Minimum: 100 MB of available space for application installation
- Additional space for cached maps and routes, depending on campus size (typically 500 MB to 1 GB)

Display:

- Minimum: 4.7-inch screen with 720p resolution
- Recommended: 5.5-inch screen or larger with 1080p resolution or higher

GPS:

- Built-in GPS receiver for real-time location tracking
- Support for GLONASS and Galileo (optional but recommended for improved accuracy).



3. SYSTEM DESIGN

3.1 ARCHITECTURE DESIGN



3.2 FLOW CHART

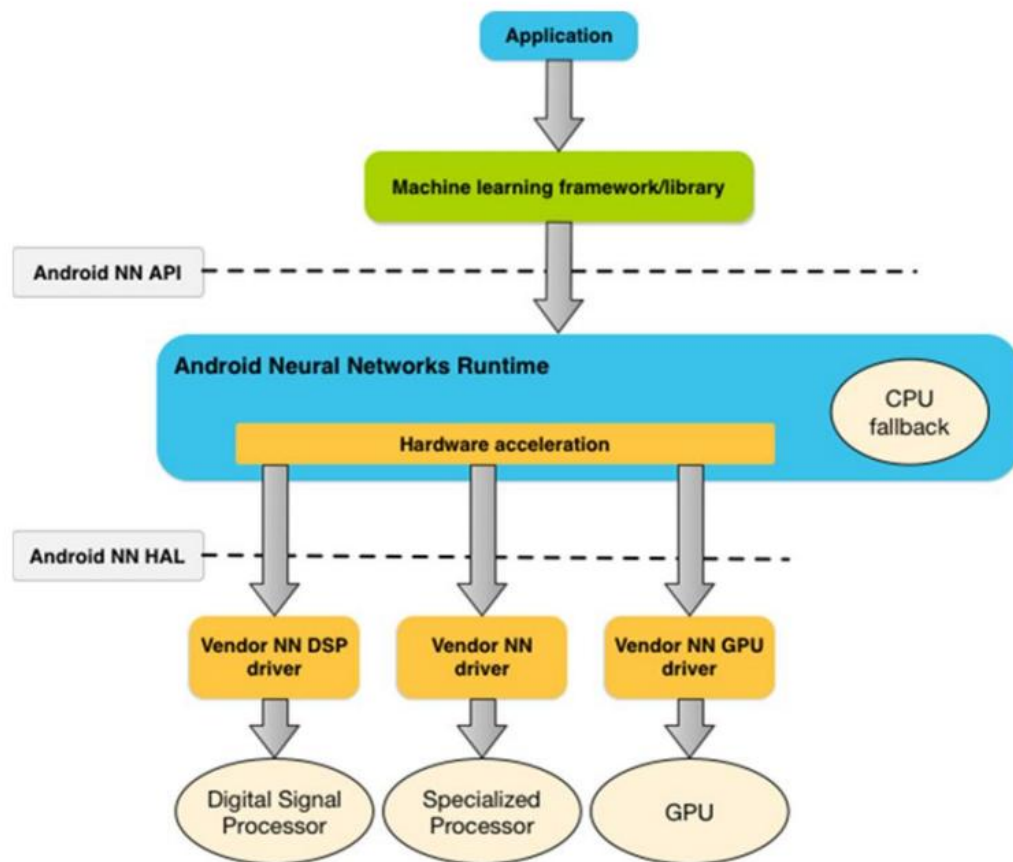


Fig 3.2 Flow Chart

3.3 MINDMAP DIAGRAM

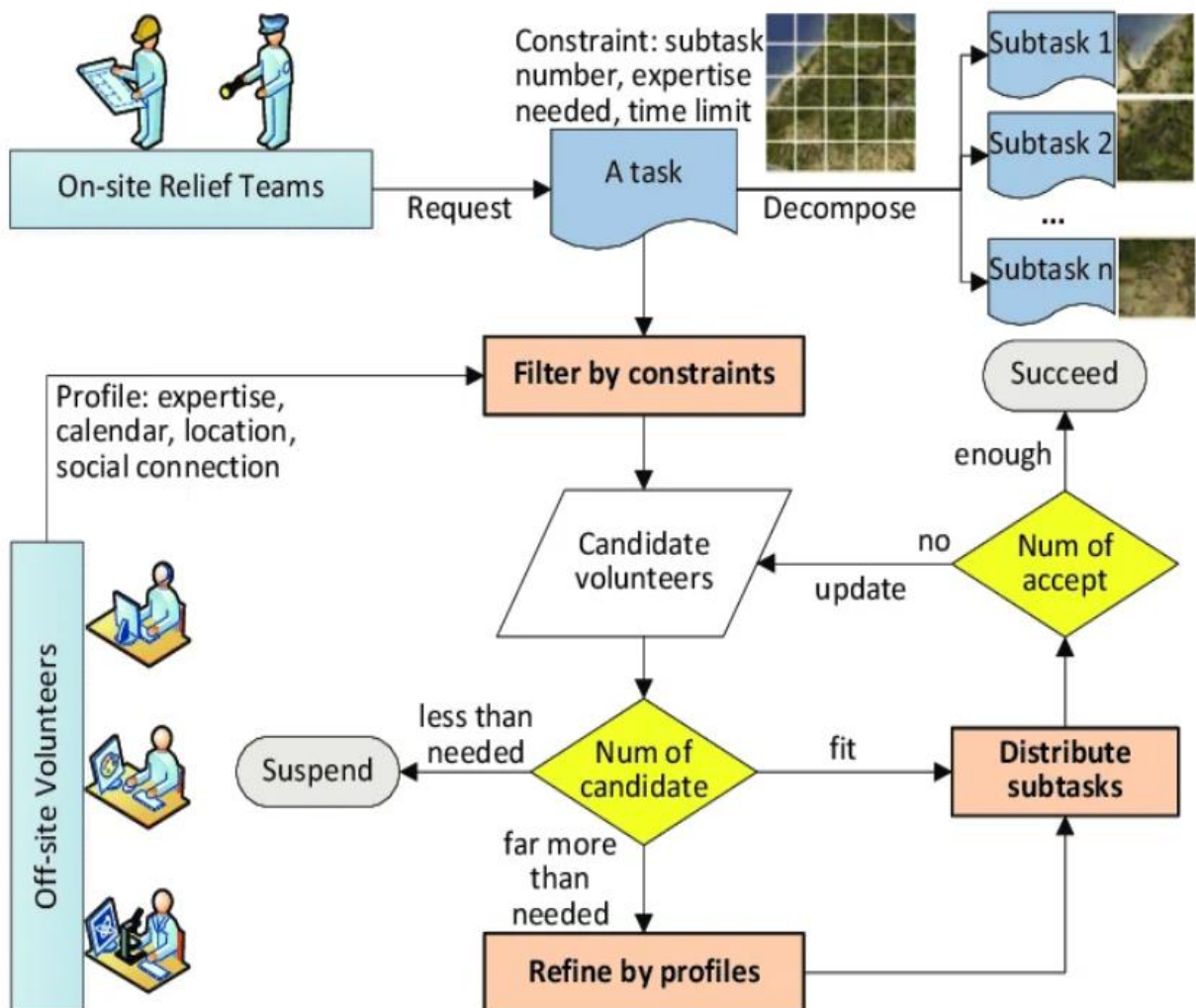


Fig 3.3 MindMap Diagram

4. CODE IMPLEMENTATION

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Crowdsourced Reporting System</title>
  <style>
    body { font-family: Arial, sans-serif; background: #f4f4f9; margin: 20px; }
    h1 { text-align: center; }
    form, .complaint, #loginSection, #logoutSection, #filterSection {
      background: #fff; padding: 20px; margin: 15px auto;
      border-radius: 8px; max-width: 500px;
      box-shadow: 0 2px 5px rgba(0,0,0,0.1);
    }
    input, textarea, select, button {
      width: 100%; padding: 10px; margin: 8px 0;
      border: 1px solid #ccc; border-radius: 5px;
    }
    button { background: #007bff; color: #fff; border: none; cursor: pointer; }
    button:hover { background: #0056b3; }
    .status { font-weight: bold; padding: 5px 10px; border-radius: 5px; display: inline-block; }
    .pending { background: #ffc107; color: #fff; }
    .working { background: #17a2b8; color: #fff; }
    .resolved { background: #28a745; color: #fff; }
  </style>
</head>
<body>

  <h1>Crowdsourced Reporting System</h1>

  <!-- Login Section -->
  <div id="loginSection">
    <h3>Login</h3>
    <input type="text" id="username" placeholder="Username" required>
    <input type="password" id="password" placeholder="Password" required>
    <button onclick="login()">Login</button>
  </div>

  <!-- Logout Section -->
  <div id="logoutSection" style="display:none;">
    <button onclick="logout()">Logout</button>
  </div>

  <!-- Category Filter (for both users and admins) -->
  <div id="filterSection" style="display:none;">
    <h3>Filter Complaints by Category</h3>
    <select id="filterCategory" onchange="displayComplaints()">
      <option value="All">All</option>
```

```

    <option value="Civic">Civic</option>
    <option value="Technical">Technical</option>
    <option value="Social">Social</option>
</select>
</div>

<!-- Complaint Form (Users only) -->
<form id="complaintForm" style="display:none;">
    <h3>Submit Complaint</h3>
    <input type="text" id="title" placeholder="Complaint Title" required>
    <textarea id="description" placeholder="Describe the issue..." required></textarea>
    <select id="category" required>
        <option value="">Select Category</option>
        <option value="Civic">Civic</option>
        <option value="Technical">Technical</option>
        <option value="Social">Social</option>
    </select>
    <button type="submit">Submit Complaint</button>
</form>

<!-- Complaint List -->
<div id="complaintsList" style="display:none;"></div>

<script>
    let currentRole = null;

    // Load complaints from localStorage
    function getComplaints() {
        return JSON.parse(localStorage.getItem("complaints")) || [];
    }

    // Save complaints to localStorage
    function saveComplaints(complaints) {
        localStorage.setItem("complaints", JSON.stringify(complaints));
    }

    // Login system
    function login() {
        const user = document.getElementById("username").value;
        const pass = document.getElementById("password").value;

        if (user === "admin" && pass === "admin123") {
            currentRole = "admin";
            alert("Logged in as Admin");
            document.getElementById("complaintForm").style.display = "none";
        } else {
            currentRole = "user";
            alert("Logged in as User");
            document.getElementById("complaintForm").style.display = "block";
        }
    }

```

```

document.getElementById("complaintsList").style.display = "block";
document.getElementById("logoutSection").style.display = "block";
document.getElementById("filterSection").style.display = "block";
document.getElementById("loginSection").style.display = "none";

displayComplaints();
}

// Logout system
function logout() {
  currentRole = null;
  document.getElementById("loginSection").style.display = "block";
  document.getElementById("complaintForm").style.display = "none";
  document.getElementById("complaintsList").style.display = "none";
  document.getElementById("logoutSection").style.display = "none";
  document.getElementById("filterSection").style.display = "none";
}

// Complaint submission (users)
document.getElementById("complaintForm").addEventListener("submit", function(e) {
  e.preventDefault();
  const complaints = getComplaints();
  const complaint = {
    id: complaints.length + 1,
    title: document.getElementById("title").value,
    description: document.getElementById("description").value,
    category: document.getElementById("category").value,
    status: "Pending"
  };
  complaints.push(complaint);
  saveComplaints(complaints);
  displayComplaints();
  this.reset();
});

// Display complaints with category filter
function displayComplaints() {
  const complaints = getComplaints();
  const list = document.getElementById("complaintsList");
  list.innerHTML = "";

  if (complaints.length === 0) {
    list.innerHTML = "<p>No complaints submitted yet.</p>";
    return;
  }

  let selectedCategory = document.getElementById("filterCategory").value;

  complaints.forEach(c => {
    if (selectedCategory === "All" || c.category === selectedCategory) {
      const div = document.createElement("div");

```

```

div.className = "complaint";
div.innerHTML = `
  <h3>${c.title}</h3>
  <p>${c.description}</p>
  <p><strong>Category:</strong> ${c.category}</p>
  <span class="status ${c.status.toLowerCase()} ">${c.status}</span>
  ${currentRole === "admin" ? `
    <br><br>
    <button onclick="updateStatus(${c.id}, 'Working')">Mark Working</button>
    <button onclick="updateStatus(${c.id}, 'Resolved')">Mark Resolved</button>
    <button onclick="deleteComplaint(${c.id})" style="background:red;">Delete</button>
  ` : ""}
`;
list.appendChild(div);
}
});
}

```

```

// Update status (admins only)
function updateStatus(id, newStatus) {
  if (currentRole !== "admin") {
    alert("Only admins can update status!");
    return;
  }
  const complaints = getComplaints();
  const complaint = complaints.find(c => c.id === id);
  if (complaint) {
    complaint.status = newStatus;
    saveComplaints(complaints);
    displayComplaints();
  }
}

```

```

// Delete complaint (admins only)
function deleteComplaint(id) {
  if (currentRole !== "admin") {
    alert("Only admins can delete complaints!");
    return;
  }
  let complaints = getComplaints();
  complaints = complaints.filter(c => c.id !== id);
  saveComplaints(complaints);
  displayComplaints();
}
</script>

```

```

</body>
</html>

```


5. RESULTS

Crowdsourced Reporting System

Complaint Title

Describe the issue...

Select Category



Submit Complaint

Fig 5.1 Complaint page

overflow of garbage

garbage is overflowing near the road (dry and wet waste)

Social



Submit Complaint

Fig 5.2 Garbage overflow

overflow of dustbin

garbage is overflowing near the road (dry and wet waste)

Category: Social

Pending

Mark Working

Mark Resolved

Fig 5.3 Did not solve yet

overflow of dustbin

garbage is overflowing near the road (dry and wet waste)

Category: Social

Working

Mark Working

Mark Resolved

Fig 5.4 Under Working

overflow of dustbin

garbage is overflowing near the road (dry and wet waste)

Category: Social

Resolved

Mark Working

Mark Resolved

Fig 5.5 Problem Resolved

6. CONCLUSION

The Crowd-Sourced Civics Issue Reporting and Resolution System provides an effective solution to the limitations of traditional civic complaint methods. By leveraging digital technology, it enables citizens to report issues quickly and accurately while ensuring transparency through real-time tracking. The inclusion of crowdsourcing improves the reliability and prioritization of complaints, allowing authorities to focus on the most critical problems.

Furthermore, the use of modern technologies like AI and geolocation enhances efficiency, reduces response time, and streamlines the resolution process. Overall, the system strengthens the connection between citizens and government, promotes accountability, and contributes to the development of smarter and more responsive cities.

7. FUTURE SCOPE

The future scope of the Crowd-Sourced Civics Issue Reporting and Resolution System is vast, with significant potential for enhancement through advanced technologies. The system can be improved by integrating Artificial Intelligence (AI) and Machine Learning (ML) to automatically detect, classify, and predict civic issues in high-risk areas. A dedicated mobile application with features like real-time notifications, multilingual support, and voice-based reporting can make the platform more accessible to a wider range of users. Additionally, the integration of IoT-based smart sensors can enable automatic detection of problems such as waste levels, water leakage, and air pollution. Introducing reward mechanisms can further encourage citizen participation and engagement. In the long run, the system can be expanded to rural regions and integrated with various government services, transforming it into a comprehensive and efficient e-governance platform.

8. REFERENCES

1. [Crowdsourced Civic Issue Reporting System \(IJERT\)](#)
2. [A Research Paper on Crowdsourced Civic Issue Reporting and Resolution System \(Zenodo\)](#)
3. [Crowdsourced Civic Issue Reporting and Resolution System \(IJPREMS\)](#)

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ABSTRACT

Rapid urbanization has led to a significant rise in civic problems such as potholes, garbage accumulation, water leakage, and damaged infrastructure. Traditional complaint management systems are often slow, inefficient, and lack transparency, resulting in delayed responses and reduced citizen satisfaction.

The Crowd-Sourced Civics Issue Reporting and Resolution System is a web- or mobile-based platform designed to empower citizens to actively participate in governance by reporting civic issues in real time. Users can submit complaints along with images, location details, and descriptions, ensuring accurate and actionable information for authorities.

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24R11A62D5	T. NavyaSri	

24R11A62D9 V. Navya

System.

PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
✓	✓	✓		✓	✓	✓	✓	✓		✓	✓	✓	

✓

✓ ✓

SDG1	SDG2	SDG3	SDG4	SDG5	SDG6	SDG7	SDG8	SDG9	SDG10	SDG11	SDG12					
					✓			✓		✓						

Batch No:		Name of the Title
Roll No	Name	
24R11A62D1		

Batch No.	Name of the Title	Nature of Project-II			
		Product	Application	Research	Others (please specify)

Table 3: Domain of the DT/Project/ Internship work (Please tick √ Appropriate for your Title)

Batch No.	Name of the Title	Domain of the Title				
		ARTIFICIAL INTELLIGENCE, MACHINE LEARNING AND DEEP LEARNING, NATURAL LANGUAGE PROCESSING (NLP)	COMPUTER NETWORKS, INFORMATION SECURITY, CYBER SECURITY, DIGITAL FORENSICS, BLOCKCHAIN TECHNOLOGY	DATA WAREHOUSING, DATA MINING, BIG DATA ANALYTICS, FULL STACK & MOBILE APPLICATION DEVELOPMENT	CLOUD COMPUTING, DEVOPS, INTERNET OF THINGS, NETWORK SYSTEMS & WIRELESS TECHNOLOGIES	SOFTWARE ENGINEERING, IMAGE PROCESSING & COMPUTER VISION

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